

VantageNav: Active Multi-View Instance Verification and Goal-Viewpoint Alignment for Zero-Shot Instance ImageGoal Navigation

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Abstract—Instance ImageGoal Navigation requires locating a specific object instance specified by a goal image in an unknown environment. Existing methods relying on observations acquired during standard exploration are vulnerable to unfavorable viewpoints and degraded visual matching, while conventional success criteria do not ensure recovery of the goal-image viewpoint. We propose VantageNav, a zero-shot framework combining active multi-view instance verification with goal-viewpoint alignment. For candidates supported by stable evidence, VantageNav actively acquires reachable complementary views and adaptively invokes box-constrained, box-free full-frame, or temporal-stability verification. Cross-view correspondences estimate the relative goal-viewpoint pose; estimates passing geometric quality verification are transformed and locked as a fixed world-frame terminal goal. On 1,000 HM3D InstanceImageNav validation episodes, VantageNav achieves 72.0% SR, 31.2% SPL, and 52.52% View Success under a $0.25\text{ m}/10^\circ$ criterion. All real-world quadruped trials succeed, with 53.2% average SPL and 13.5m average travel distance.

I. INTRODUCTION

Instance ImageGoal Navigation (IIN) specifies a target object instance using a goal image [1]. Unlike navigation with semantic-category goals [2], [3], IIN must both locate the target in an unknown environment and verify that observations from different viewpoints correspond to the specified instance. Reliable discovery and verification therefore remain challenging under limited observations, viewpoint changes, occlusion, and degraded visual matching [4], [5].

Existing IIN methods improve reliability through target matching, candidate verification, and terminal localization. Local matching and candidate verification strengthen recognition, but target observations are still acquired mainly during standard exploration [4], [6]. Reconstruction-based methods generate additional views [7], [8], yet these are primarily virtual renderings rather than online observations acquired through physical motion; active viewpoint selection has also been studied mainly for offline pre-collected instance verification [9]. Moreover, candidate discovery and verification often share the same visual pipeline, so degraded detection, region extraction, or local matching can prevent the true target from reaching later verification stages [4], [6]. Although candidate verification and false-positive rejection have been explored [10], [11], adaptive use



Fig. 1. Real-world VantageNav trials on the Unitree GO2. Each row shows the goal image and representative observations along the navigation trajectory until successful target localization.

of complementary verification routes with a common geometric quality test remains underexplored. Recent work has addressed goal-image pose recovery and precise terminal localization [12], [13], [14], [15], [16], but converting a locally estimated goal-viewpoint pose into a fixed world-frame terminal pose and preserving it during replanning has received less systematic attention.

To address these issues, we propose VantageNav, a zero-shot IIN framework for active multi-view observation and goal-viewpoint alignment. Our contributions are:

1) **Candidate-driven active complementary multi-view observation:** To address the limited target viewpoints available during exploration, VantageNav generates and filters reachable, viewpoint-complementary observation poses around a candidate target using the online map, and actively navigates to the selected viewpoints to acquire additional RGB-D observations, improving discrimination among visually similar instances.

2) **Evidence-driven multi-route target verification:** To handle missed detections, bounding-box errors, and degraded local matching, VantageNav adaptively selects box-constrained region, box-free full-frame, or temporal-stability verification according to the available

visual evidence, with all routes unified by geometric quality verification for final target confirmation.

3) **Cross-view relative pose estimation and goal-viewpoint alignment:** Using dense cross-view correspondences, VantageNav estimates the relative goal-viewpoint pose. Estimates passing geometric quality verification are transformed and locked as a fixed world-frame SE(2) navigation goal, after which path tracking and terminal control align both the goal-viewpoint position and heading.

On 1,000 HM3D validation episodes, VantageNav achieves 72.00% SR, 31.20% SPL, and 52.52% View Success under a 0.25 m/10° terminal-alignment criterion. Real-world experiments in multiple scenes further validate its effectiveness and practicality.

II. RELATED WORK

A. Target Perception and Multi-View Observation

Existing ImageNav methods use end-to-end policies, topological graphs, or visual memory to search for targets using observations collected during exploration [17], [18]. In IIN, Mod-IIN [6] uses local feature matching, while IEVE [4] approaches detected candidates to improve matching. GaussNav [7] and SplatSearch [8] generate novel views from 3D reconstruction, but mainly through virtual rendering. Active recognition methods accumulate visual evidence through controlled motion and further consider active viewpoint selection for candidate verification [19]. However, how to combine online mapping with the active acquisition of reachable, viewpoint-complementary observations remains underexplored in IIN. VantageNav addresses this gap by physically acquiring such complementary observations for zero-shot instance verification.

B. Image-Goal Matching and Instance Verification

Visual matching and candidate verification are central to instance-level recognition. Mod-IIN [6] and IEVE [4] use local matching and distance-dependent keypoint thresholds; CrossIA [5] reduces the reference-observation domain gap through contrastive learning. UniGoal [10], BSC-Nav [20], and Context-Nav [11] respectively use scene-graph verification, structured spatial memory, and viewpoint-aware 3D constraints. Because a primary visual pipeline may still miss the true target under detection or matching degradation, VantageNav uses three complementary verification routes followed by unified geometric quality verification: the routes reduce missed detections under different visual conditions, while geometric verification suppresses false confirmations.

C. Goal-Pose Estimation and Terminal Alignment

Some ImageNav methods map matches to topological nodes or map locations for local navigation [6], [18]. SLING [12] improves last-mile navigation, RNR-Map [13] supports image localization and camera tracking, IGL-Nav [14] recovers target-view poses with incremental 3D

Gaussians, and AnyImageNav [15] estimates the goal-image 6-DoF camera pose from dense geometry. MG-Nav [16] combines spatial memory, geometric alignment, and local image-goal control, while GeniNav [21] selects trajectories using semantic and geometric constraints. In contrast, VantageNav focuses on stable post-verification execution by transforming the estimated planar relative goal-viewpoint pose into a fixed world-frame SE(2) goal, preserving it during subsequent replanning, and using independent terminal control for final position and heading alignment.

III. PROBLEM FORMULATION

We follow standard Instance Image-Goal Navigation (IIN) [1] and additionally require the agent to recover the observation position and orientation associated with the goal image. The agent uses egocentric RGB-D and odometry relative to the start pose, with no prior map or target location.

IV. METHODOLOGY

Fig. 2 summarizes VantageNav. Goal-category inference and LLM-based similar-category expansion support candidate detection, while full-frame and candidate-region similarities update global and local value maps for exploration or complementary viewpoint generation (Sec. IV-A/B). Additional observations are processed by box-constrained region, box-free full-frame, or temporal-stability verification followed by geometric quality verification (Sec. IV-C). A verified relative goal-viewpoint pose is then transformed into a fixed world-frame SE(2) goal for path tracking, replanning, and terminal heading alignment (Sec. IV-D).

A. Goal-Conditioned Candidate Construction and Value-Map Update

Given goal image I_g , the system infers its primary category c_g . To reduce category errors caused by viewpoint changes, occlusion, and intra-class similarity, an LLM generates visually confusable categories $\mathcal{C}_{\text{sim}}$, yielding $\mathcal{C}_g = \{c_g\} \cup \mathcal{C}_{\text{sim}}$. This set is used only for candidate detection and instance segmentation; final verification relies on visual, temporal, and geometric evidence.

For current RGB observation I_t , full-frame similarity s_t^{full} is mapped to the observed region as

$$V_t^{\text{global}}(u) = \text{clip}(w_{\text{full}} s_t^{\text{full}}, 0, 1). \quad (1)$$

Here, u is a map cell and w_{full} the full-frame similarity weight. This value measures regional target relevance but does not determine instance identity.

For candidate region j , region similarity $s_{j,t}^{\text{crop}}$ is computed against the goal image. Valid depth points are transformed to the world frame using depth and odometry, and spatially associated observations are fused into

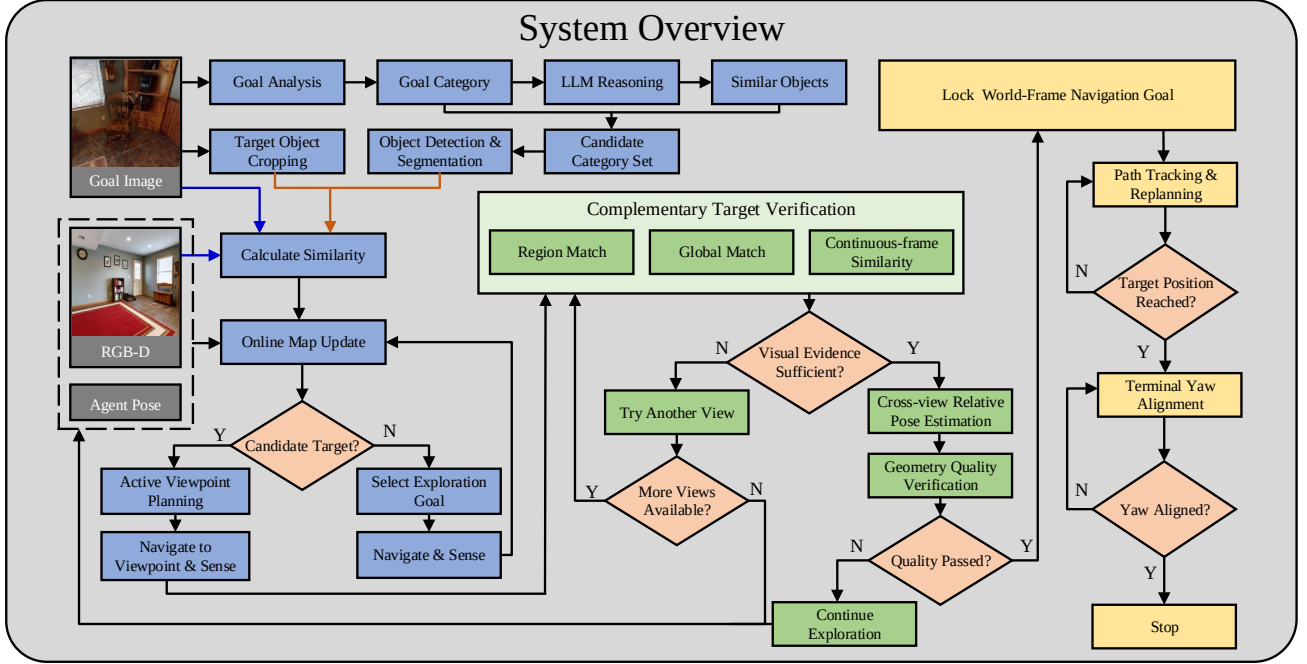


Fig. 2. Overview of VantageNav. Goal-conditioned exploration identifies candidate instances, active viewpoint planning acquires complementary observations, and multi-route visual verification is followed by geometric quality verification. Verified relative poses are converted into a fixed world-frame goal viewpoint for path tracking, replanning, and terminal heading alignment.

the same candidate. With previous accumulated evidence $S_{j,t-1}$ and valid-observation count $n_{j,t-1}$,

$$S_{j,t} = \frac{n_{j,t-1}S_{j,t-1} + s_{j,t}^{\text{crop}}}{n_{j,t-1} + 1}. \quad (2)$$

Thus, $S_{j,t}$ is the cumulative average similarity over $n_{j,t}$ valid region observations and is used for semantic-map fusion and candidate ranking.

Persistent evidence is represented by consecutive high-score length $h_{j,t}$, recent evidence $r_{j,t}$, and evidence-cycle index $e_{j,t}$. The consecutive count is

$$h_{j,t} = \begin{cases} h_{j,t-1} + 1, & \text{if } j \text{ is re-associated and } s_{j,t}^{\text{crop}} \geq \tau_s, \\ 0, & \text{otherwise.} \end{cases} \quad (3)$$

Here, $r_{j,t}$ is the mean similarity of the two latest consecutive high-score observations and becomes valid after two such observations; $e_{j,t}$ distinguishes new from processed evidence. Failed candidates are subject to cooldown or attempt limits to avoid repeated checks without new evidence.

The active-observation candidate set is

$$\mathcal{I}_t = \left\{ j \mid \begin{array}{ll} h_{j,t} \geq H_{\min}, & r_{j,t} \geq \tau_r, \\ e_{j,t} > e_j^{\text{done}}, & j \notin \mathcal{K}_t \end{array} \right\}. \quad (4)$$

$H_{\min}$ is the minimum consecutive high-score length, τ_r is the recent-evidence threshold, e_j^{done} is the last processed evidence cycle, and $\mathcal{K}_t$ contains candidates under cooldown or suppression. Full-frame similarity therefore

guides where to explore, while consecutive stability, recent evidence, and evidence novelty determine which instance warrants active inspection.

B. Candidate-Driven Active Complementary Multi-View Observation

A candidate satisfying the active-observation condition may still be ambiguous under occlusion, viewpoint bias, or missing texture. VantageNav therefore generates reachable complementary viewpoints from candidate geometry and the online map, as shown in Fig. 3.

The fused candidate depth cloud is projected to the 2D plane, with its median as a geometric anchor. Viewing directions are sampled from the principal direction of the candidate point cloud and neighboring directions. For direction d and distance r ,

$$v_j(d, r) = p_{j,d}^{\text{surf}} + rd, \quad (5)$$

where $p_{j,d}^{\text{surf}}$ is the front-surface point along d . This construction keeps the viewpoints consistent with object geometry rather than uniformly sampling around the candidate.

Generated viewpoints are checked for map bounds, free space, obstacle inflation, target visibility, and A* reachability. Each feasible viewpoint is scored using visible-surface ratio $\rho(v)$, FOV compatibility $g_{\text{fov}}(v)$, direction consistency $b(v)$, unknown-space ratio $\eta(v)$,

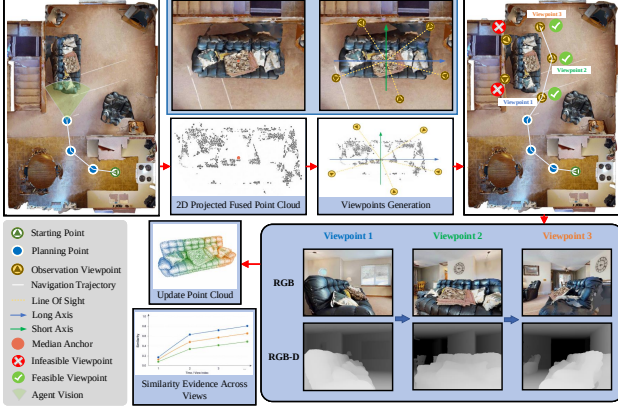


Fig. 3. Candidate-driven active multi-view observation. Candidate geometry is projected onto the ground plane to generate complementary observation viewpoints, which are filtered by free-space, visibility, and reachability constraints. The robot sequentially visits selected viewpoints to acquire additional RGB-D evidence and update the candidate point cloud.

current-position distance, and path length $L(v)$:

$$J(v) = w_1\rho(v) + w_2g_{fov}(v) + w_3b(v) + w_4\mathbb{I}[\phi_{\min} \leq \phi(v) \leq \phi_{\max}] - w_5\|v - p_t\|_2 - w_6\eta(v) - w_7L(v). \quad (6)$$

Here, p_t is the current robot position, $\phi(v)$ the candidate's position within the FOV, and $w_1, \dots, w_7$ fixed weights. The score favors visible, well-framed, complementary views while penalizing distance, unknown space, and path cost. Feasible viewpoints are ranked by $J(v)$, spatially de-duplicated, and visited sequentially to update RGB-D evidence and the candidate point cloud.

C. Evidence-Driven Multi-Route Target Verification

Active observations may still contain missed detections, box offsets, or degraded local features. VantaNav therefore uses box-constrained region, detector-independent full-frame, and temporal-stability verification, with all triggered observations passed to the same geometric quality verification stage (Fig. 4).

Using region semantic similarity $s_{j,t}^{\text{crop}}$ from Sec. IV-A, define full-frame and crop-level sparse match counts as $G_t = |\mathcal{M}(I_g, I_t)|$ and $C_{j,t} = |\mathcal{M}(I_g^{\text{crop}}, I_{j,t}^{\text{crop}})|$, where $\mathcal{M}$ is the sparse matching operator. Algorithm 1 summarizes route triggering and arbitration.

Route 1 uses both full-frame and candidate-region matches, Route 2 preserves verification when the current detection box is unreliable, and Route 3 exploits semantic similarity and temporal box consistency. A triggered route only initiates geometric verification; final target verification and pose estimation still depend on the shared geometric quality test.

Algorithm 1 Evidence-Driven Multi-Route Verification Trigger

Require: $G_t, \mathcal{C}_g, (N^-, B^-, y^-), \{C_{j,t}, s_{j,t}^{\text{crop}}\}_{j \in \mathcal{D}_t}, \{y_{j,t}, B_{j,t}\}_{j \in \mathcal{D}_t}, \text{ thresholds}$

Ensure: $\text{route}_t, (N_t, B^+, y^+)$

- 1: $b_1 \leftarrow \text{false}; N_t \leftarrow 0; B^+, y^+ \leftarrow \emptyset$
- 2: **if** $\mathcal{D}_t \neq \emptyset \wedge G_t \geq \tau_G$ **then** ▷ R1
- 3: $j_t^{\text{match}} \leftarrow \arg \max_{j \in \mathcal{D}_t} C_{j,t}$
- 4: $b_1 \leftarrow (y_{j_t^{\text{match}}, t} \in \mathcal{C}_g) \wedge (C_{j_t^{\text{match}}, t} \geq \tau_C)$
- 5: **end if**
- 6: $b_2 \leftarrow \neg b_1 \wedge (G_t \geq \tau_{\text{nobox}})$ ▷ R2
- 7: $\mathcal{D}_t^g \leftarrow \{j \in \mathcal{D}_t \mid y_{j,t} \in \mathcal{C}_g\}$
- 8: **if** $\mathcal{D}_t^g \neq \emptyset$ **then** ▷ R3
- 9: $j_t^{\text{sem}} \leftarrow \arg \max_{j \in \mathcal{D}_t^g} s_{j,t}^{\text{crop}}$
- 10: $q_t \leftarrow \text{isfinite}(s_{j_t^{\text{sem}}, t}^{\text{crop}}) \wedge (s_{j_t^{\text{sem}}, t}^{\text{crop}} \geq \tau_T)$
- 11: **if** q_t **then**
- 12: $a_t \leftarrow (B^- \neq \emptyset) \wedge (y^- = y_{j_t^{\text{sem}}, t}) \wedge (\text{IoU}(B^-, B_{j_t^{\text{sem}}, t}) \geq \tau_{\text{IoU}})$
- 13: $N_t \leftarrow N^- + 1$ **if** a_t **else** 1
- 14: $(B^+, y^+) \leftarrow (B_{j_t^{\text{sem}}, t}, y_{j_t^{\text{sem}}, t})$
- 15: **end if**
- 16: **end if**
- 17: $\text{route}_t \leftarrow \text{none}$
- 18: **if** b_1 **then**
- 19: $\text{route}_t \leftarrow \text{R1}$
- 20: **else if** b_2 **then**
- 21: $\text{route}_t \leftarrow \text{R2}$
- 22: **else if** $N_t \geq K_{\min}$ **then**
- 23: $\text{route}_t \leftarrow \text{R3}$
- 24: **end if**
- 25: **return** $\text{route}_t, N_t, B^+, y^+$

D. Cross-View Relative Pose Estimation and Goal-Viewpoint Alignment

Once verification is triggered, dense cross-view correspondences between the goal image and the current RGB-D observation, together with depth, are used to estimate the current-to-goal relative viewpoint pose. In the robot frame, it is represented by the forward offset e_f , lateral offset e_l , and yaw increment e_ψ , as shown in Fig. 5.

A unified geometric quality gate rejects estimates degraded by weak texture, depth noise, or mismatches. Let N_m , N_v , and N_i be the total, confidence-valid, and rigid-registration inlier match counts, respectively, with inlier ratio $\gamma = N_i / \max(1, N_v)$. Let e_d be 3D depth-consistency error and $e_p = \sqrt{e_f^2 + e_l^2}$ the planar displacement. The estimate is valid when

$$Q_t = \mathbb{I} \left[\begin{array}{l} N_m \geq \bar{N}_m, \quad N_v \geq \bar{N}_v, \quad N_i \geq \bar{N}_i, \\ \frac{N_i}{\max(1, N_v)} \geq \gamma_{\min}, \quad e_d \leq \epsilon_d, \\ e_p \leq \epsilon_p, \quad |e_\psi| \leq \epsilon_\psi \end{array} \right]. \quad (7)$$

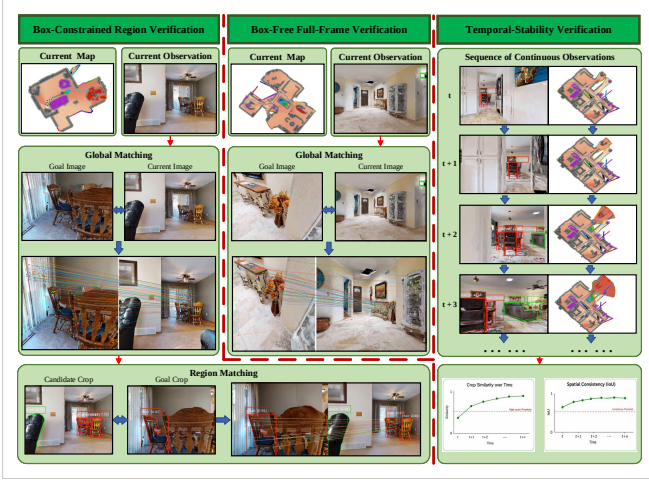


Fig. 4. Evidence-driven multi-route target verification. Box-constrained region verification, box-free full-frame verification, and temporal-stability verification provide complementary evidence under detection or matching degradation. All triggered routes are followed by the same geometric quality verification stage before final target confirmation.

Estimates with $Q_t = 0$ are discarded; those with $Q_t = 1$ are used to construct the terminal goal-viewpoint pose. Threshold values are given in the implementation details.

Given the current world-frame robot pose $g_t = (x_t^w, y_t^w, \psi_t)$, the verified local pose (e_f, e_l, e_ψ) is transformed into the fixed goal-viewpoint pose $g_\star = (x_\star^w, y_\star^w, \psi_\star)$:

$$\begin{aligned} x_\star^w &= x_t^w + e_f \cos \psi_t - e_l \sin \psi_t, \\ y_\star^w &= y_t^w + e_f \sin \psi_t + e_l \cos \psi_t, \\ \psi_\star &= \text{wrap}(\psi_t + e_\psi). \end{aligned} \quad (8)$$

Here, $g_\star$ is the terminal observation pose required to recover the goal-image viewpoint, not the object’s 3D pose. Once generated, it is locked to prevent frame-to-frame matching fluctuations from shifting the terminal goal.

A* replans on the updated occupancy map toward the same $g_{\star,xy}$; map changes, path invalidation, or collisions therefore change only the path, not the locked terminal goal. After reaching its vicinity, translation stops and terminal yaw alignment is performed. Alignment is complete when $\|p_k - g_{\star,xy}\|_2 \leq \delta_{\text{dist}}$ and $|\text{wrap}(\psi_k - \psi_\star)| \leq \delta_{\text{yaw}}$.

V. EXPERIMENTS

A. Simulation Experimental Setup

Task and Dataset: We evaluate VantageNav in Habitat [22] on the HM3D InstanceImageNav v3 validation set [23], containing 36 scenes, 1,000 episodes, and 795 target instances.

Evaluation Metrics: We report success rate (SR), success weighted by path length (SPL), goal-viewpoint position error e_{pos} , yaw error e_{yaw} , and View Success. View Success requires final errors no greater than 0.25 m

and 10° ; these thresholds are used only for evaluation, not internal stopping.

Implementation Details: The agent uses six discrete actions with 0.25 m translation and 30° rotation, a 640×480 RGB-D camera, and a 1,000-step limit. YOLOv7 [24] and GroundingDINO [25] generate candidates; DINO [26] computes semantic similarity; DISK [27] and LightGlue [28] perform cross-view matching; Qwen3-8B [29] generates similar categories; and MAST3R [30] supports geometric quality verification. Navigation uses A* and rolling path tracking. Parameters are: $w_{\text{full}} = 0.45$, $\tau_s = 0.45$, $H_{\text{min}} = 2$, $\tau_r = 0.50$ (Sec. IV-A); $r \in \{1.40, 1.70, 2.00, 2.30\}$ m, $\rho_{\text{min}} = 0.55$, $(w_1, \dots, w_7) = (3, 2, 0.5, 0.25, 0.10, 0.35, 0.15)$, $\phi_{\text{min}} = 25^\circ$, $\phi_{\text{max}} = 65^\circ$ (Sec. IV-B); $\tau_G = 100$, $\tau_C = 10$, $\tau_{\text{nobox}} = 150$, $\tau_T = 0.60$, $\tau_{\text{IoU}} = 0.20$, $K_{\text{min}} = 2$ (Sec. IV-C); $(\bar{N}_m, \bar{N}_v, \bar{N}_i, \gamma_{\text{min}}) = (60, 40, 25, 0.30)$, $(\epsilon_d, \epsilon_p, \epsilon_\psi) = (0.35 \text{ m}, 4.0 \text{ m}, 120^\circ)$, $\delta_{\text{dist}} = 0.20 \text{ m}$, $\delta_{\text{yaw}} = 15^\circ$ (Sec. IV-D). All parameters are fixed before validation evaluation.

Baselines: RL Baseline [1] uses task-specific end-to-end RL. ZSON [31] and OVRL-v2-IIN [17] use multi-modal goal representations; PSL [32] strengthens candidate reasoning through prioritized semantic learning, and GOAT [33] uses open-vocabulary goal understanding. Mod-IIN [6] decouples goal understanding, exploration, and navigation without task-specific training; IEVE [4] combines semantic exploration and instance verification; UniGoal [10] uses a unified goal representation; and T²-Nav [34] improves exploration through temporal graph memory and loop detection.

B. Comparison with State-of-the-art

Table I compares VantageNav with existing methods on HM3D InstanceImageNav. Without task-specific training, VantageNav achieves 72.0% SR and 31.2% SPL. Among all compared methods, it attains the second-highest SR, only 0.6 percentage points below T²-Nav (72.6%), while achieving the highest SPL and outperforming T²-Nav (27.8%) by 3.4 percentage points. Compared with other training-free methods such as Mod-IIN and UniGoal, VantageNav also achieves clear improvements in both SR and SPL. These results indicate that VantageNav maintains competitive navigation success

TABLE I
COMPARISON WITH EXISTING METHODS ON HM3D
INSTANCEIMAGENAV.

Method	Training-Free	SR↑	SPL↑
RL Baseline [1]	No	8.3	3.5
ZSON [31]	No	14.6	7.3
OVRL-v2-IIN [17]	No	24.8	11.8
PSL [32]	No	23.0	11.4
GOAT [33]	No	37.4	16.1
IEVE (InternImage) [4]	No	70.2	25.2
Mod-IIN [6]	Yes	56.1	23.3
UniGoal [10]	Yes	60.2	23.7
T ² -Nav [34]	Yes	72.6	27.8
VantageNav (Ours)	Yes	72.0	31.2

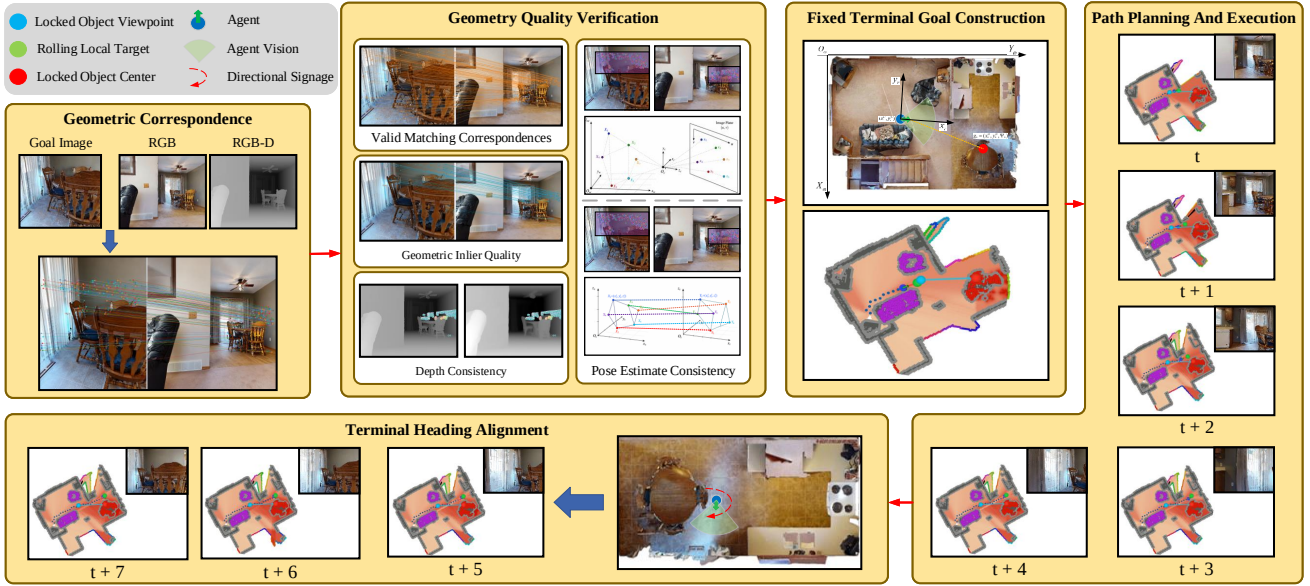


Fig. 5. Cross-view relative pose estimation and goal-viewpoint alignment. Cross-view correspondences and depth estimate the relative goal-viewpoint pose, which is accepted after geometric quality verification and converted into a fixed world-frame SE(2) goal. Subsequent replanning changes only the path, while the locked terminal goal is preserved for final position and yaw alignment.

while improving path efficiency.

C. Failure Cause Analysis

Task outcomes over the 1,000 episodes are categorized as:

- **Success (A):** Habitat returns success and task conditions are met.
- **False Negative (B):** Trajectory enters within 1 m of the target but is not judged successful.
- **No Frontier (C):** Verification incomplete and no traversable frontier remains.
- **Stepout (D):** 1,000-step limit reached before success.
- **Infeasible (E):** Height-band pre-screening finds no same-floor target and terminates early.
- **Stucking (F):** Agent persistently fails to produce valid motion.
- **False Positive (G):** Target-stop branch triggered but Habitat returns failure.

As shown in Fig. 6, *False Negative* is the dominant failure mode (11.9%), indicating that the agent may reach the target vicinity yet fail to complete verification because of unstable sparse matching, incorrect instance association, or unmet terminal conditions, preventing subsequent dense geometric estimation. Insufficient evidence can subsequently lead to frontier exhaustion or step-limit termination.

No Frontier mainly reflects occupancy-map and frontier-filtering limitations, whereas large scenes, long exploration, and repeated verification increase the risk of *Stepout*. *Infeasible* mainly results from height-representation bias in floor determination, and *Stucking*

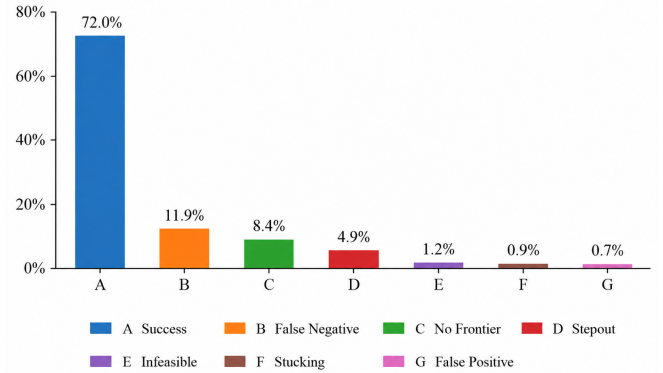


Fig. 6. Distribution of success and failure causes for VantageNav over 1,000 HM3D InstanceImageNav episodes.

from collisions, blocked local paths, or map-traversability mismatch. The relatively low *False Positive* rate suggests that false stopping is not the main bottleneck and that sparse matching plus dense geometry suppress incorrect stopping to some extent. Overall, target verification and exploration efficiency remain the main limitations.

D. Ablation Study

As shown in Table II, removing Active Inspection reduces SR, SPL, and View Success, confirming the benefit of actively acquired complementary observations. Removing any verification route also degrades performance; the largest SR drop, from 72.0% to 61.5%, occurs without box-free full-frame verification, showing its role in recovering from missed detections. The box-constrained and temporal-stability routes provide complementary gains.

TABLE II
ABLATION STUDY OF VANTAGENAV ON HM3D INSTANCEIMAGENAV.

Configuration	Navigation Performance			Pose Precision [†]	
	SR (%) ↑	SPL (%) ↑	View Success (%) ↑	e_{pos} (m) ↓	e_{yaw} (°) ↓
VantageNav	72.0	31.20	52.52	0.1732	5.77
w/o Active Inspection	64.6	29.51	46.30	0.1800	5.65
w/o Box-Constrained Region Verification	70.0	30.00	52.30	0.1747	5.82
w/o Box-Free Full-Frame Verification	61.5	27.21	46.00	0.1721	5.32
w/o Temporal-Stability Verification	70.4	29.81	52.30	0.1732	5.63
w/o Viewpoint-Guided Terminal Control	51.2	23.04	1.10	0.9692	35.57

[†] Pose errors are averaged over successful episodes only; successful subsets may differ across configurations.

Because e_{pos} and e_{yaw} are averaged only over successful episodes, successful subsets differ across configurations. Except for Viewpoint-Guided Terminal Control, pose errors therefore remain similar and small variations should not be interpreted as improved terminal accuracy. Removing Viewpoint-Guided Terminal Control increases e_{pos} from 0.1732 m to 0.9692 m and e_{yaw} from 5.77° to 35.57°, while View Success falls from 52.52% to 1.10%. Thus, this module determines not only terminal heading but also final approach position and stopping time, making it critical to both navigation success and goal-viewpoint recovery.

E. Real-world Deployment

Real-world experiments use a Unitree GO2 quadruped with an NVIDIA GeForce RTX 4060 Laptop GPU and 12th Gen Intel Core i7-12800HX ROS host, an Intel RealSense D435i depth camera, and a 4D LADAR L2; Point-LIO [35] provides odometry. Kinodynamic A* performs path search, GCopter based on MINCO [36] generates smooth trajectories, and MPC tracks them through the Unitree GO2 SDK. The camera height is 0.41 m. We conduct four real-world navigation trials, as shown in Fig. 1; all succeed, with overall and per-trial results reported in Table III and Table IV, respectively.

The four trials achieve 53.2% average SPL and 13.50 m average travel distance. Episode 2 has a 25° terminal yaw error, likely because the visible target and scene

geometry weakly constrain orientation: cross-view correspondences still estimate position relatively stably but discriminate vertical-axis rotation less reliably. Smaller yaw errors in the remaining trials indicate that goal-viewpoint accuracy depends on both position estimation and geometric orientation constraints.

VI. CONCLUSIONS

We present VantageNav, a zero-shot IIN framework combining active multi-view observation with goal-viewpoint alignment. Complementary observations, multi-route verification, and geometric quality verification improve instance reliability, while the verified relative pose is transformed into a fixed world-frame terminal goal for replanning and final heading control. On 1,000 HM3D v3 validation episodes, VantageNav achieves 72.0% SR, 31.2% SPL, and 52.52% View Success; all real-world trials succeed with 53.2% average SPL and 13.5 m average travel distance.

Although locking the terminal goal suppresses drift across consecutive observations, it cannot correct an initially biased pose; active multi-view inspection also depends on reachable viewpoints with sufficient target visibility, making fully occluded or interaction-dependent targets challenging. Future work will investigate multi-view information fusion, adaptive target verification, and active multi-view observation under occlusion.

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TABLE III

OVERALL REAL-WORLD NAVIGATION PERFORMANCE. RESULTS ARE AVERAGED OVER 4 TRIALS. AVG.TD: AVERAGE TRAVELED DISTANCE; AVG.TT: AVERAGE TRAVELED TIME.

Trials	SR (%)	SPL (%)	Avg. TD (m)	Avg. TT (s)
4	100	53.2	13.50	158

TABLE IV

REAL-WORLD NAVIGATION RESULTS FOR INDIVIDUAL EPISODES. TD: TRAVELED DISTANCE; TT: TRAVELED TIME; e_{pos} : TERMINAL POSITION ERROR; e_{yaw} : TERMINAL YAW ERROR.

Episode	TD (m)	SPL	TT (s)	e_{pos} (m)	e_{yaw} (°)
Episode 1	14.30	77.90%	140	0.54	10
Episode 2	18.05	36.29%	253	0.37	25
Episode 3	13.02	54.04%	153	0.16	6
Episode 4	8.61	44.65%	87	0.85	5

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